

LIST OF MINORS

1a.	AEROSPACE ENGINEERING Old UGARC (Y21 and earlier)	MINOR		Template No.AE
		Title	AEROSPACE ENGINEEIRNG	
		C O U R S E S		
			Compulsory Course:	
			AE201A [05]	
			Any THREE from:	
			AE211A [11]	
			AE321A [09]	
			AE331A [09]	
		AE650A [09] / AE753A [09] / AE341 [11]		
32-36				

1b.	AEROSPACE ENGINEERING New UGARC (Y22 and later)	MINOR		Template No.AE
		Title	AEROSPACE ENGINEEIRNG	
		C O U R S E S		
			Compulsory Course:	
			AE201M [05]	
			Any THREE from:	
			AE211 [9]	
			AE321 [9]	
			AE333 [09]	
		AE341 [11]		
32-34				

LIST OF MINORS

2a.	BIOLOGICAL SCIENCES AND BIO-ENGINEERING Old UGARC (Y21 and earlier)	MINOR		Template No. BSBE	
		Title	TISSUE ENGINEERING		
		C O U R S E S	BSE211A [09]		
			BSE411A [11]		
			BSE421A [11]		
			-		
		31			

2b.	BIOLOGICAL SCIENCES AND BIO-ENGINEERING New UGARC (Y22 and later)	MINOR		Template No. BSBE	
		Title	TISSUE ENGINEERING	ENGINEERING IN MEDICINE	
				Any three courses	
		C O U R S E S	BSE211 [9]	BSE616 [9]	
			BSE411 [9]	BSE644 [9]	
			BSE421 [9]	BSE656 [9]	
				BSE662 [9]	
		-			
		27			

LIST OF MINORS

3a.	CHEMICAL ENGINEERING Old UGARC (Y21 and earlier)	MINOR		Template No. CHE	
		Title	CHEMICAL ENGINEERING		
		C O U R S E S	CHE251A [09]		
			CHE261A [09]		
			CHE313A [09]		
			CHE331A [09]		
			-		
			36		

3b.	CHEMICAL ENGINEERING New UGARC (Y22 and later)	MINOR		Template No. CHE	
		Title	CHEMICAL ENGINEERING		
		Any three courses			
		C O U R S E S	CHE251		
			CHE212		
			CHE213		
			CHE221		
			CHE311		
			CHE331		
CHE381					
CHE352					

LIST OF MINORS

4.	CHEMISTRY	MINOR		Template No. CHM-5		
		Title	PHYSICAL CHEMISTRY*	INORGANIC CHEMISTRY	ORGANIC CHEMISTRY	
			Compulsory Courses:	Compulsory Courses:	Compulsory Courses:	
		C O U R S E S	CHM321A [09]	CHM345A [09]	CHM201A [09]**	
			CHM322A [09]	CHM342A [09]	CHM202A [09]	
			Any TWO from:	Any TWO from:	CHM303A [09]	
					CHM402A [09]	
					-	
			CHM621A [09] CHM622A [09] CHM626A [09] CHM636A [09] CHM637A [09] CHM650A [09] CHM664A [09] CHM685A [09] CHM689A [09] CHM695A [09] CHM696A [09] CHM699A [09]	CHM616A [09] CHM631A [09] CHM646A [09] CHM647A [09] CHM651A [09] CHM654A [09] CHM668A [09] CHM691A [09]		
				36	36	36
			REMARKS:			
			1) *CSO202A is a pre-requisite for Minor in Physical Chemistry			
2) **EXCEPTION: CHM201A to be done by Y13 and onward batches. Y11 and Y12 batch shall do CHM203A. Course title, content and credits are same. Just the course number has been changed.						

LIST OF MINORS

5.	CIVIL ENGINEERING	MINOR		Template No. CE-5		Remark: 1. Minor in Infrastructure Engineering and Management under Civil Engineering department has been discontinued.
		Title	Infrastructure Engineering and Management			
		C O U R S E S	CE241 [09]	[odd semester]		
			CE640 [09]	[odd semester]		
			CE641A [05]	[odd semester]		
			CE643A [05] or CE644A [05]	[even semester]		
			CE667A [05]	[even semester]		

LIST OF MINORS

6.	COMPUTER SCIENCE & ENGINEERING	MINOR			Template No. CSE-4	
		ALGORITHMS	COMPUTER SYSTEMS	THEORY OF COMPUTING	ARTIFICIAL INTELLIGENCE* ¹	MACHINE LEARNING & APPLICATIONS %
		COMPULSORY COURSE(S):	COMPULSORY COURSE(S):	COMPULSORY COURSE(S):	COMPULSORY COURSE(S):	COMPULSORY COURSE(S):
		CS210A [12]/ ESO207A[12]	CS210A [12]/ ESO207A[12]	CS210A [12]/ ESO207A[12]	CS210A [12]/ ESO207A [12]	CS210A [12]/ ESO207A [12]
		CS345A [09]		CS340A [09]	CS365A [09]	CS771A [09]
		And, ONE from:	And, TWO from:	And, ONE from:	And, ONE from:	And, ONE from:
		CS645A [09]	CS220A [12]	CS640A [09]	CS671A [09]	CS365A [09]
		CS646A [09]	CS330A [12]	CS641A [09]	CS672A [09]	CS671A [09]
		CS647A [09]	CS315A [09]	CS642A [09]	CS674A [09]	CS676A [09]
		CS648A [09]	CS335A [12]	CS643A [09]	CS675A [09]	CS678A [09]
		CS663A [09]	CS422A [09]	CS644A [09]	CS676A [09]	CS685A [09]
		CS664A [09]	CS425A [09]	CS649A [09]	CS685A [09]	CS698I [09]
		CS719A [09]	CS433A [09]	CS680A [09]	CS686A [09]	CS698N [09]
		CS743A [09]	CS455A [09]	CS681A [09]	CS782A [09]	CS698S [09]
				CS687A [09]		CS772A [09]
				CS740A [09]		CS773A [09]
				CS741A [09]		CS774A [09]
				CS742A [09]		-
	% Additional rules to obtain Minor in Machine Learning and Applications (ML): if a student of Y22 and Y23 batch has already done a course (December 19, 2025), equivalent to CS771 course (currently, ten courses i.e., AE646, CE784, CHE657, EE656, EE798Z, MSE643, PHY654, DMS673, EE708, ME644.), may claim Minor in Machine Learning and Applications (ML) with the following conditions: i. Mandatory ESO207 ii. Two courses from the approved AI/ML course basket (available in the Minor's template) iii. One course from outside the department that is considered equivalent to CS771. (Currently, ten courses are identified equivalent to CS771 i.e., AE646, CE784, CHE657, EE656, EE798Z, MSE643, PHY654, DMS673, EE708, ME644.)	Remarks: 1) For each Minor stream, the minimum total credit requirement is 30. 2) In each minor stream any related course(s), even if not mentioned in the list of optional courses, may be taken as optional with the permission of the CSE DUGC. 3) *1 = Only for students graduating till 2017.				

LIST OF MINORS

7.	ELECTRICAL ENGINEERING	MINOR				Template No.EE-4	
		Title	MICROELECTRONICS/ DIGITAL SYSTEMS	POWER	COMMUNICATIONS & SIGNAL PROCESSING	CONTROLS	RF & PHOTONICS
		C O U R S E S	EE200A [11]	EE200A [11]	EE200A [11]	EE200A [11]	EE200A [11]
			EE210A [11]	EE330A [11]	EE320A [11]	EE250A [11]	EE340A [11]
			EE311A [09]/ EE370A [11]	EE360A [09]	EE321A [09]/ EE301A [09]	EE650A [09]	EE612A [09]/ EE642A [09]/ EE648A [09]
			31/33	31	31	31	31

Old UGARC (Y21 and earlier): courses with suffix "A"

New UGARC (Y22 and later): courses without suffix "A"

LIST OF MINORS

8.	HUMANITIES & SOCIAL SCIENCES (New Template)	MINOR				Template No. HSS-1
		Title	LINGUISTICS	ENGLISH LITERATURE	PHILOSOPHY	
		C O U R S E S	Pre-requisite: Completion of a Level-I course in Linguistics with at least a B grade.	Pre-requisite: Completion of any one Literature course with at least a B grade.	Pre-requisite: Completion of any one Philosophy course.	
			ONE course from following:	THREE courses from following:		
			ENG122A [11] ENG124A [11]	ENG123A [11] ENG125A [11] ENG431A [09] ENG432A [09] ENG433A [09] ENG434A [09] ENG435A [09] ENG436A [09] ENG437A [09] ENG438A [09] ENG439A [09] ENG440A [09] ENG445A [09] ENG446A [09] ENG456A [09] ENG457A [09] ENG458A [09] ENG460A [09] ENG466A [09] HSS405A [09] ENG719A [09] ENG733A [09] ENG743A [09] ENG744A [09] ENG750A [09] ENG 752 [09] ENG 753 [09] ENG 764 [09]	PHI141 PHI448 PHI147 PHI452 PHI142 PHI455 PHI143 PHI443M PHI145 PHI757 PHI446 PHI774 PHI447 PHI140 PHI161 PHI450 PHI481 PHI453 PHI441 PHI445	
			and, TWO courses from following:			
			ENG401 [09] ENG402 [09] ENG404 [09] ENG408 [09] ENG423 [09] ENG443 [09] ENG447 [09] ENG448 [09] ENG452 [09] ENG467 [09]			
			Total Required Credits for Minor: 29			
				Total Required Credits for Minor: 27-31		
					Total minimum Required Credits for Minor: 27	
			Note: Courses as detailed above with the suffix 'A' will also be counted for the minor.			
Remarks:						
1.) If a student has done a course in English other than those listed above and wishes to have a Minor in English (Linguistics or English Literature), he/she may request the DUGC, HSS, to approve of the same.						
2.) For Linguistics Minor: The request of students claiming minor with ENG7XY courses may be considered by the DUGC on the case-to-case basis if a students take, these courses up to 2024-25-I semester only.						

Remarks:

- 1.) If a student has done a course in English other than those listed above and wishes to have a Minor in English (Linguistics or English Literature), he/she may request the DUGC, HSS, to approve of the same.
- 2.) For Linguistics Minor: The request of students claiming minor with ENG7XY courses may be considered by the DUGC on the case-to-case basis if a students take, these courses up to 2024-25-I semester only.

LIST OF MINORS

9.	INDUSTRIAL AND MANAGEMENT ENGINEERING	MINOR		Template No. IME-3		
		Title	INDUSTRIAL AND MANAGEMENT ENGINEERING			
		C O U R S E S	DE PG-1			
			DE PG-2			
			DE PG-3			
			-			
			27			
		<u>REMARKS:</u>				
1) DE PG may be from IME or MBA courses.						

LIST OF MINORS

10.	MATERIALS SCIENCE & ENGINEERING	MINOR		Template No.MSE-5
		Title	FUNCTIONAL MATERIALS	
		C O U R S E S	Any THREE from:	
			MSE604A [09]	
			MSE624A [09]	
			MSE628A [09]	
			MSE631A [09]	
			MSE693A [09]	
27				

LIST OF MINORS

11.	MATHEMATICS AND SCIENTIFIC COMPUTING	MINOR		Template No. BS-MTH-5	Remark: 1. Minor in Applied Mathematics under Mathematics and Scientific Computing department has been discontinued.
		Title	APPLIED MATHEMATICS		
		C O U R S E S			
			MTH308A [11]		
			MTH428A [11]		
			MTH426A [09]		
			31		

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12.	MECHANICAL ENGINEERING	MINOR		Template No. ME-5
		Title	MANUFACTURING SCIENCES	COMPUTATIONAL TECHNIQUES IN MECHANICAL ENGINEERING
	Old UGARC (Y21 and earlier): courses with suffix “A” except ME767 & ME653 New UGARC (Y22 and later): courses without suffix “A”	C O U R S E S	Any THREE from:	Any THREE from:
			ME661A [09] (Prerequisite TA202A)	ME623A [09] (Prerequisite ESO202A, MSO203B)
			ME662A [09] (Prerequisite TA202A)	ME630A [09] (Prerequisite ESO201A, ESO204A, MSO203B)
			ME663A [09] (Prerequisite ESO202A, MSO203B)	ME685A [09] (Prerequisite – No backlog in core courses)
			ME664A [09] (Prerequisite TA202A, ESO204A or Equivalent Heat Transfer Course)	ME751A [09] (Prerequisite TA101A)
			ME665A [09] (Prerequisite TA202A)	MEXXXA [09] / ME767 [09] (Nonlinear Finite Element Techniques in Solid Mechanics)
			ME774A [09] (Prerequisite ESO202A, MSO203B)	MEXXXA [09] / ME653 [09] (Atomistic Simulations in Engineering)
			ME751A [09] (Prerequisite TA101A)	-
			ME761A [09] (Prerequisite TA202A)	-
			27	27

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13a.	PHYSICS	MINOR		Template No. BS-PHY	
	Old UGARC (Y21 and earlier)	Title	STATISTICAL MECHANICS	QUANTUM MECHANICS	
		C O U R S E S			
			PSO201A [08]	PSO201A [08]	
			PHY210A [06]	PHY210A [06]	
			PHY226B [06]	PHY226B [06]	
			PHY412A [11]	PHY431A [11]	
			31	31	

13b.	PHYSICS	MINOR				Template No. BS-PHY
	New UGARC (Y22 and later)	Title	STATISTICAL MECHANICS	QUANTUM MECHANICS	GRAVITY	SOFT MATTER
		C O U R S E S				
			PSO201/PHY114	PSO201/PHY114	PHY112/PHY113/PHY115	PSO203M
			PHY211	PHY431	PHY226M	PHY412
			PHY412	PHY302/PHY626/PHY643	PHY407	PHY628
		30-33	28-33	28	28	

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14.	COGNITIVE SCIENCE	MINOR		Template No. CONG.-1	Remark: 1.Updated template of Minor in Cognitive Science was approved by the Senate in its 550 th Meeting (item No. 6). 2. The updated template is applicable from July-2022 onwards. 3. ** The project course is a semester long project that a student must take with any ONE or Jointly (more than one) of the faculty from the Department of Cognitive Science.
		Title	COGNITIVE SCIENCE		
			Compulsory courses:		
		C O U R S E S	CGS401A [09]		
			CGS402A [09]		
			and, ONE courses from following list:		
			CGS702A [09]**		
			CGS621A [09]		
			CGS622A [09]		
			CGS609A [09]		
			CGS698D [09]		
			CGS691A [09]		
			CGS671A [09]		
			CGS651A [09]		
			CS786A [09]		
			BSE656A [09]		
BSE662A [09]					
	27				

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15.	NUCLEAR ENGINEERING TECHNOLOGY	MINOR		Template No. BS-NET 2	Remark: 1. Minor in Nuclear Engineering Technology has been discontinued. Students may request for retrospective Minor if they completed the course requirements.
		Title	NUCLEAR ENGINEERING TECHNOLOGY		
		C O U R S E S			
			NT602A [09]		
			NT611A [09]		
			NT614A [09]		
		-			
	27				

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16.	STATISTICS AND DATA SCIENCE	MINOR		Template No. BS-SDS	
		Title	STATISTICS AND DATA SCIENCE		
		C O U R S E S			
			MSO205A[11] or HSO201A[11] or CS203B[5]		
			MTH211A[11]		
			MTH441A[10]		
			MTH208A[5]		
	31-37				

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17.	Sustainable Energy Engineering	MINOR		Template No. SEE-1	Remark: **As per this template, a student must take one compulsory course and a minimum 2 courses from the remaining list in which a maximum of one course is allowed from MBA courses (MBA681A, MBA683A, MBA782A). For example, a student can take either 2 SEE courses or 1 SEE course and 1 MBA course apart from the compulsory course.
		Title	Sustainable Energy Engineering		
		One Compulsory course:			
		SEE605A [09]			
		And minimum Two Department Elective Courses			
		SEE601 [09]			
		SEE602 [09]			
		SEE603 [09]			
		SEE604 [09]			
		SEE606 [09]			
		SEE607 [09]			
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		SEE619 [09]			
		SEE620 [09]			
		SEE621 [09]			
		SEE622 [09]			
		SEE623 [09]			
		SEE624 [09]			
		SEE625 [09]			
		SEE626M [09]			
		SEE627 [09]			
		SEE628 [09]			
		At most one of the following courses if taken as PG-DE:**			
		MBA681 [09]			
		MBA683 [09]			
		MBA782 [09]			
		Total Minimum Credits: 27			

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18.	Space, Planetary & Astronomical Sciences & Engineering (SPASE)	MINOR		Template No. SPASE-1	Remark: All future courses from the department, except the ones entirely related to mathematical methods or numerical techniques will be available as elective courses to complete minor.	
		Title	Space Science			
			Compulsory course:			
		C O U R S E S	SPA201			
			27 credits worth of courses from Elective courses offered by the SPASE department			
			SPA401 [09]			
			SPA402 [09]			
			SPA404 [09]			
			SPA601 [09]			
			SPA603 [09]			
			SPA611 [09]			
			SPA614M [05]			
			SPA616 [09]			
			SPA618M [05]			
			SPA621 [09]			
			SPA627M [05]			
			SPA633M [05]			
			SPA634 [10]			
			SPA636M [05]			
			SPA637 [09]			
Total minimum credits: 36						

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19.	Economics (ECO)	MINOR		Template No. ECO-1	Note: Students should also take ECO111: Economy, Society, and Public Policy as part of the SCHEME basket or as an Open Elective.
		Title	Economics		
			Compulsory courses:		
		C O U R S E S	HSO201 (11) or MSO201 (11) or MSO205 (11) or CS203 (5)		
			ECO211 (11)		
			ECO231 (11)		
			ECO251 (11)		
			Minimum 38-44 credits		

LIST OF MINORS

20.	Joint Minor from the departments: 1.) Space, Planetary and Astronomical Sciences and Engineering (SPASE) 2.) Aerospace Engineering (AE)	MINOR		Template No. SPASE and AE-1	Remark: - All four courses cannot be from one single department. - The credit of the course chosen in the joint minor must not be duplicated with either AE minor or SPASE minor. - Students from Space, Planetary and Astronomical Sciences and Engineering (SPASE) and Aerospace Engineering (AE) departments are not eligible to opt minor in these departments.
		Title	Space Technology		
			courses:		
		C O U R S E S	AE641 (9) or SPA620 (9)		
			SPA626 (9)		
			AE642 (9) or AE645 (4) or SPA637 (9)		
			AE644 (9)		
			SPA402 (9)		
			SPA601 (9)		
			AE777 (9)		
			SPA615 (9)		
			Minimum 36 credits		

The End